**Project Design Phase-II**
**Data Flow Diagram & User Stories**

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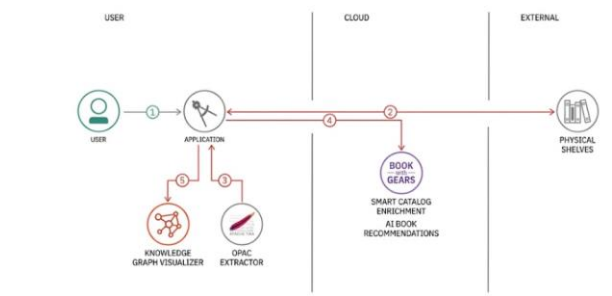
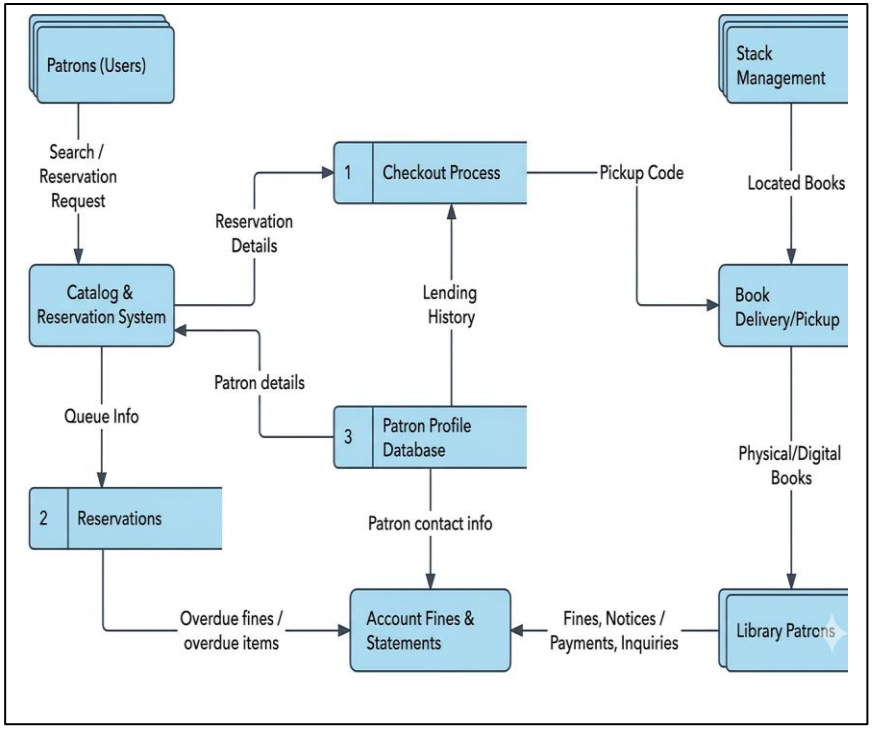
|---|---|
| Date | 05 Apr. 26 |
| Team ID | SWTID-2026-6042 |
| Project Name | Smart Library Request Workflow in ServiceNow |
| Maximum Marks | 4 Marks |

**Data Flow Diagrams:**

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.




1. User authenticates via single sign-on for library services and launches the application.
2. User selects a list of books from physical shelves (external source) to catalog and process
3. OPAC Extractor pulls catalog metadata and details for the selected books.
4. Book details are passed to Smart Catalog Enrichment for AI-driven recommendations and thematic linking.
5. Enriched book data and recommendations are visually displayed in the UI as an interactive Knowledge Graph.



## User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Customer (Mobile user) | Library Card Management | USN-LIB-1 | As a new patron, I can apply for a digital library card by entering my personal details and scanning a form of ID. | I receive a confirmation email with a temporary card number and barcode within 1 hour. | High | Sprint-1 |
| | | USN-LIB-2 | As a patron, I can view a digital version of my library card on the app and use it at self-checkout kiosks. | The digital barcode is scannable at library kiosks, and my card status (active/delinquent) is clearly visible. | High | Sprint-1 |
| | Catalog Search & Recommendations | USN-LIB-3 | As a patron, I can use a smart voice search to find books by title, author, or vague description (e.g., 'books about time travel and romance'). | The search results are accurate and display the real-time physical availability or digital download status. | Medium | Sprint-1 |
| | | USN-LIB-4 | As a patron, I can receive personalized book recommendations based on my borrowing history. | A 'Recommended for You' section appears on my dashboard, listing relevant titles with the reason for recommendation (e.g., 'Because you read Dune'). | Medium | Sprint-2 |
| | Borrowing & Returns | USN-LIB-5 | As a patron, I can reserve a book and get an in-app map showing its precise location on the shelf. | When I enter the library, the app guides me to the correct aisle and shelf with an AR overlay. | High | Sprint-3 |
| Customer (Web user) | Digital Resource Access | USN-WEB-6 | As a patron, I can access and stream high-quality audiobooks directly from the library's web portal. | The audio player works seamlessly on major web browsers with bookmarked progress saved to my account. | High | Sprint-2 |
| | | USN-WEB-7 | As a patron, I can access academic databases and research journals from home. | My library credentials provide full access to premium external databases (e.g., JSTOR, EBSCO) through a single sign-on. | High | Sprint-2 |
| | Meeting Room Booking | USN-WEB-8 | As a student group, I can book a smart meeting room on the web portal. | I can see the real-time availability of rooms, book a slot, and receive a digital key to unlock the room. | Low | Sprint-2 |
| Librarian | Collection Management | USN-STAFF-9 | As a librarian, I can use a smart inventory handheld device to instantly find and correct misplaced books. | The device provides directional cues and haptic feedback as I scan shelves, identifying off-shelf items. | High | Sprint-1 |
| | Patron Service | USN-STAFF-10 | As a librarian, I can view an unified patron profile including their preference and current issues to provide better service. | A single view shows their digital and physical history, feedback, and any open support tickets. | Medium | Sprint-1 |
| Administrator | System Configuration | USN-ADMIN-11 | As an administrator, I can configure the global library fine policies and lending duration settings. | A secure admin panel allows for editing and saving policy parameters, with immediate effect on user accounts. | High | Sprint-1 |
| | Data Analytics | USN-ADMIN-12 | As an administrator, I can view an interactive dashboard with real-time analytics on library usage and popular resources. | Visualizations of foot traffic, popular book topics, and database usage are clear and can be filtered by date. | Low | Sprint-2 |